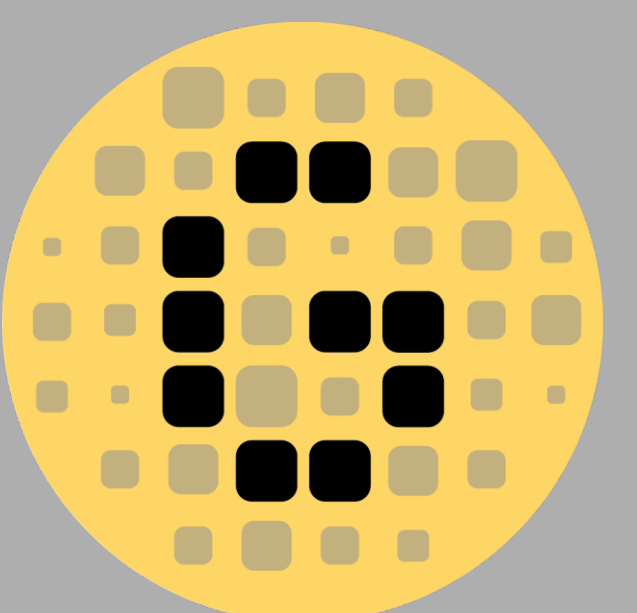


Fast PSF Synthesis with Defocus and Spherical Aberration

Nicholas Ganino and Qi Guo

Purdue University

<https://hankel.qiguo.org>



Elmore Family School of Electrical and Computer Engineering

Guo Lab @ Purdue ECE

Motivation

Real-Time Depth of Field Rendering

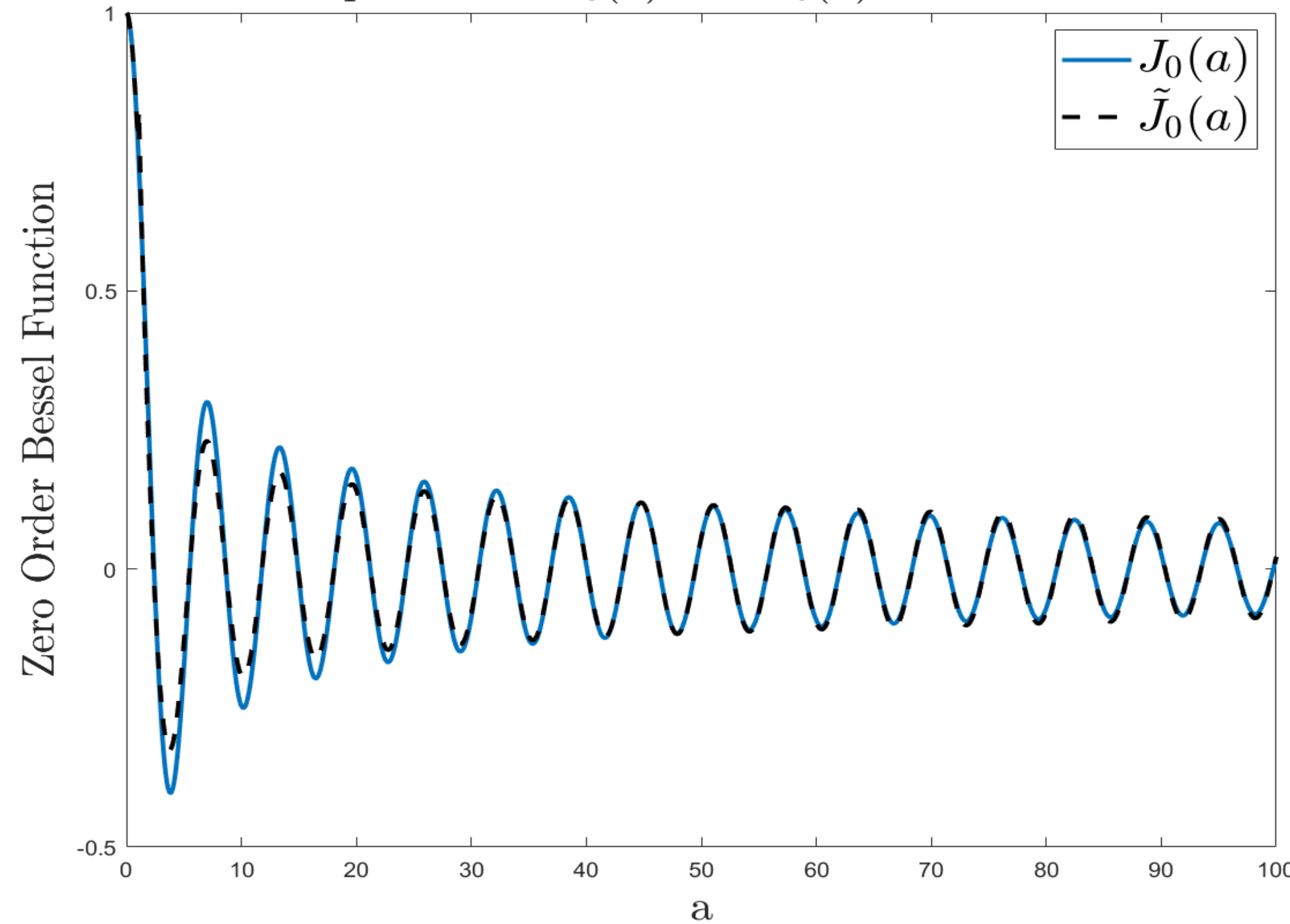


Bessel Function Approximation

$$\tilde{J}_0(a) = \begin{cases} 1 - \frac{a^2}{4} + \frac{a^4}{64}, & a \leq 1 \\ \sqrt{\frac{2}{\pi}} \left(\frac{3}{2} \alpha^{-\frac{1}{2}} - \frac{a}{2} \alpha^{-\frac{3}{2}} + \frac{1}{a} \right) \cos \left(a - \frac{\pi}{4} \right), & a > 1 \end{cases}$$

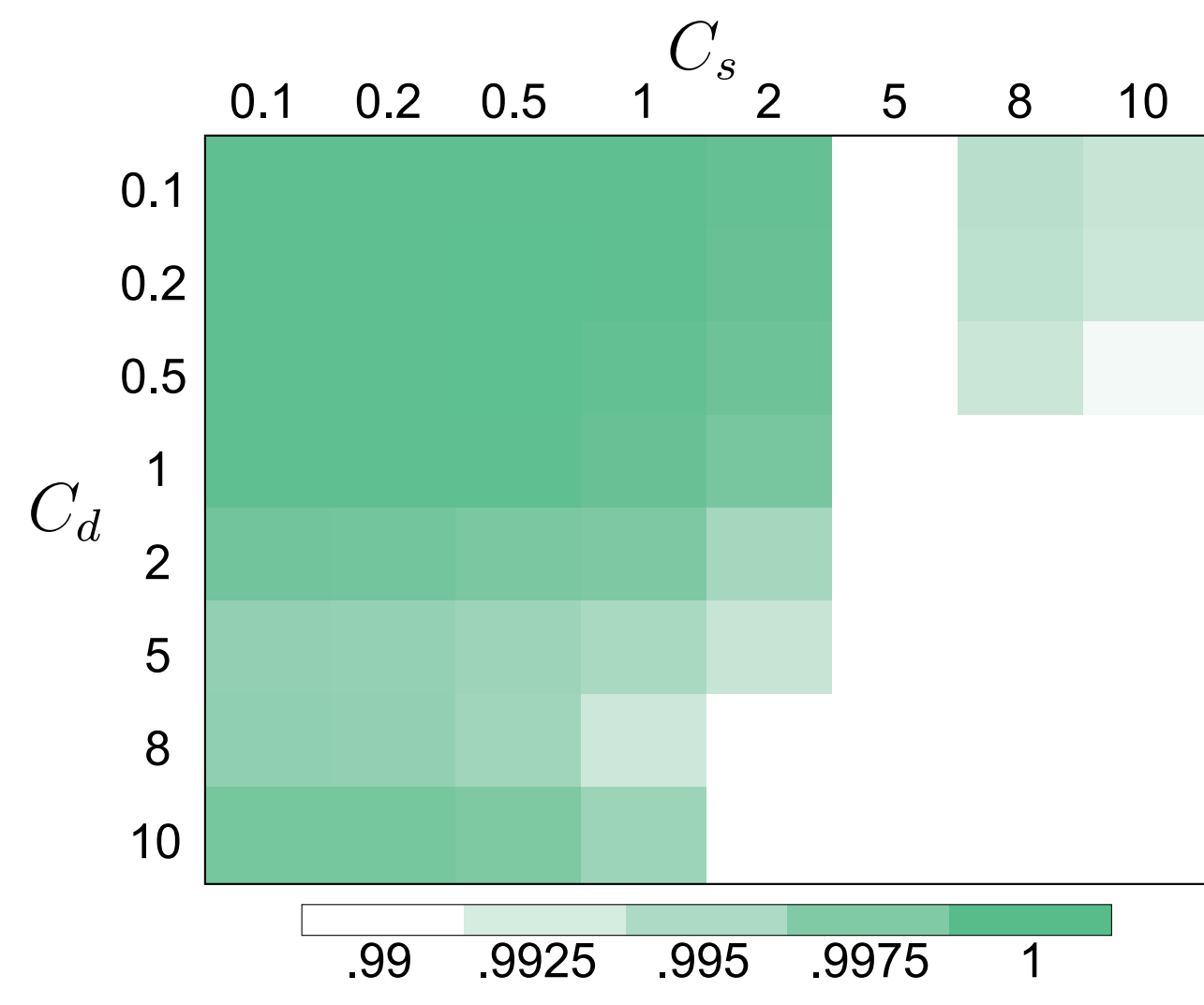
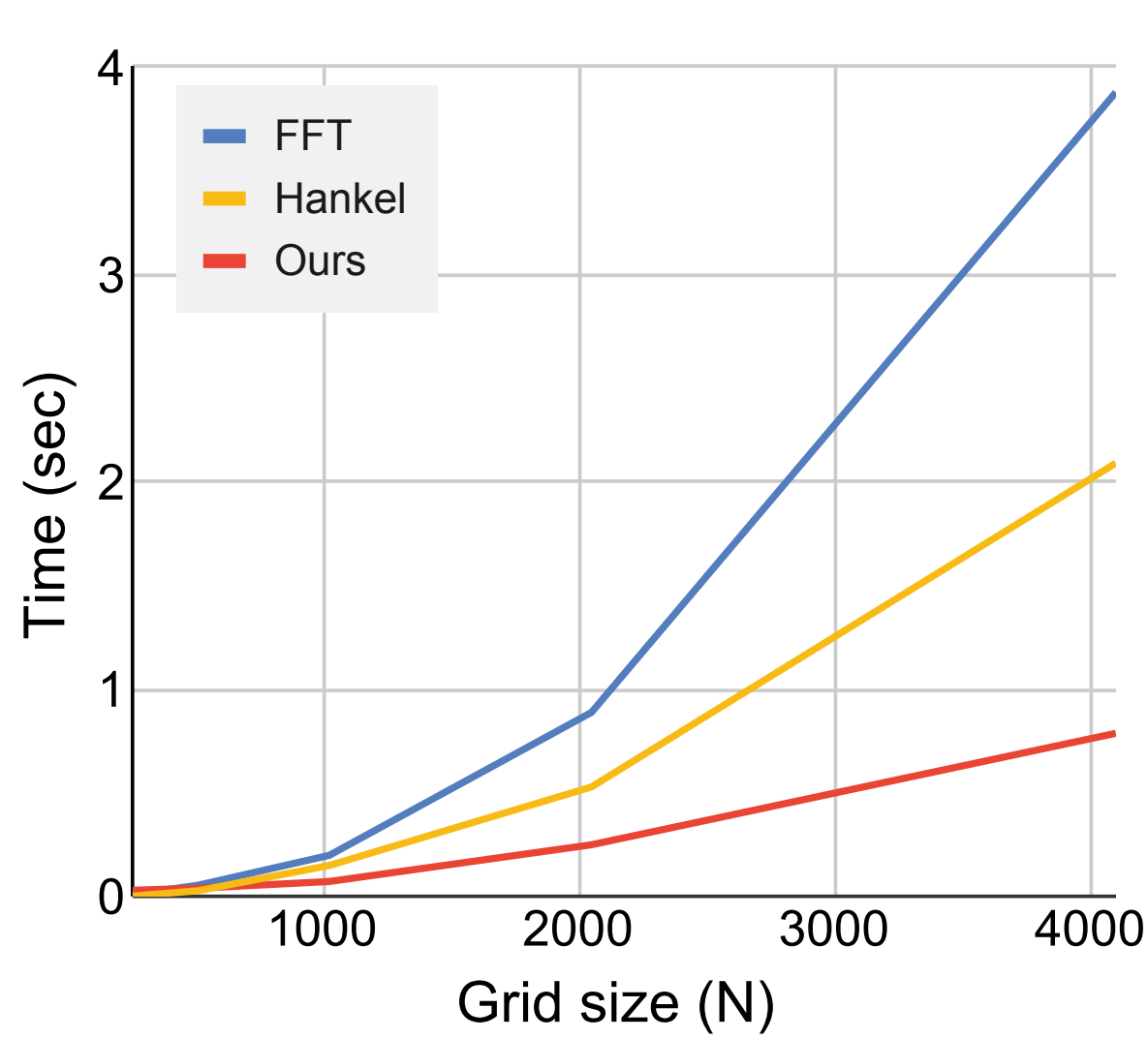
Hyperparameter α controls approximation midpoint.

Comparison of $J_0(a)$ and $\tilde{J}_0(a)$ with $\alpha = 100$



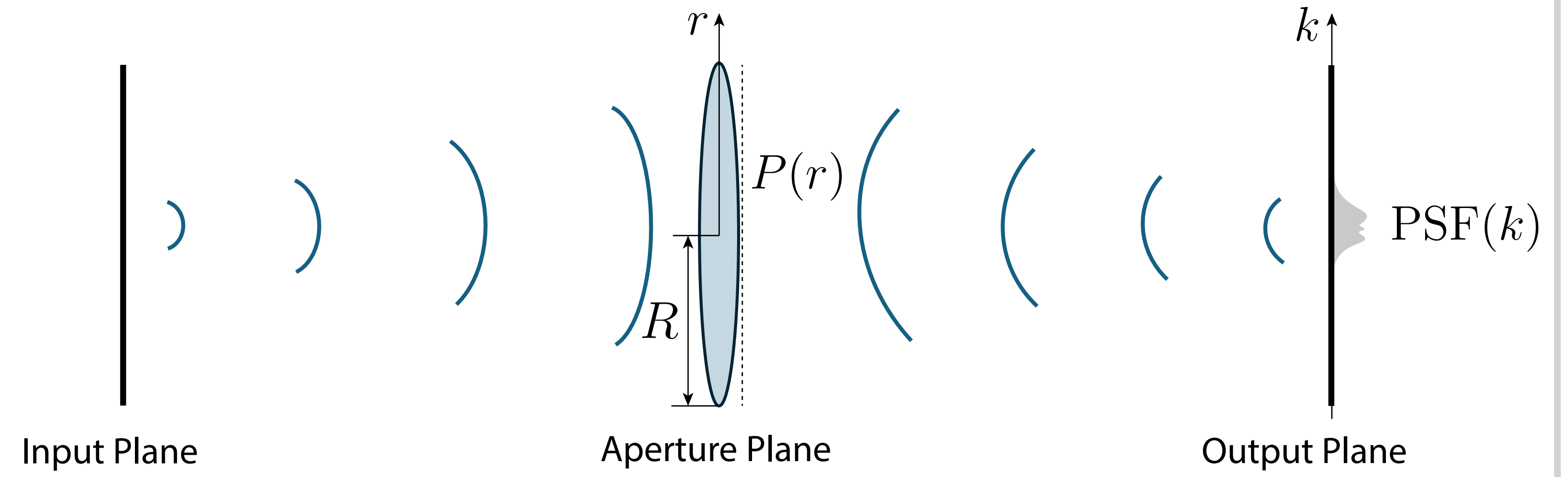
Main Contribution. The approximation provides a closed-form solution to 1st order Hankel Transform.

Computational Speed



Our closed-form solution scales better than direct Hankel transform or FFT, while being accurate with $C_s < 1$.

System Overview



First Order Hankel Transform

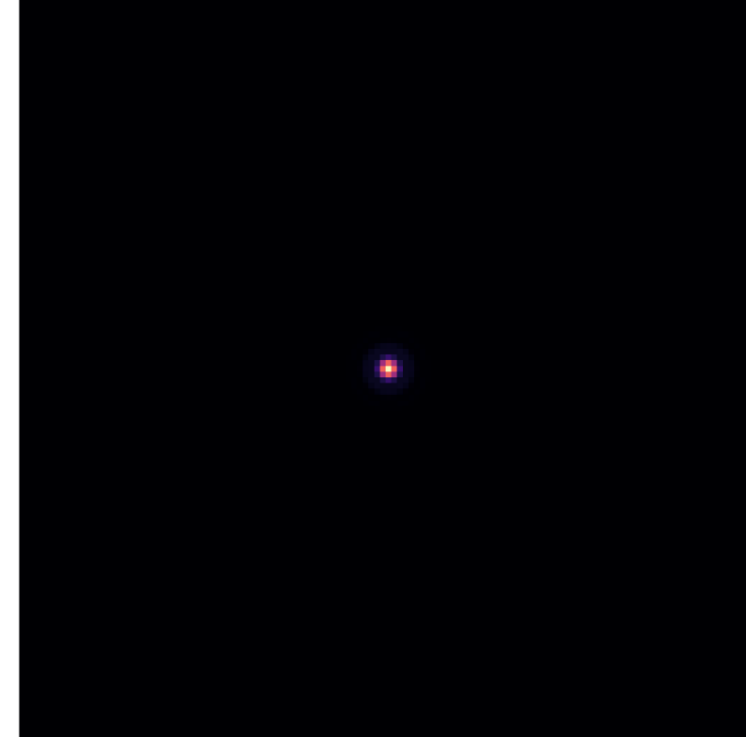
$$\text{PSF}(k) = \left| 2\pi \int_0^R P(r) J_0(2\pi k r) r dr \right|^2 \approx \left| 2\pi \int_0^R P(r) \tilde{J}_0(2\pi k r) r dr \right|^2$$

Supported Pupil Function

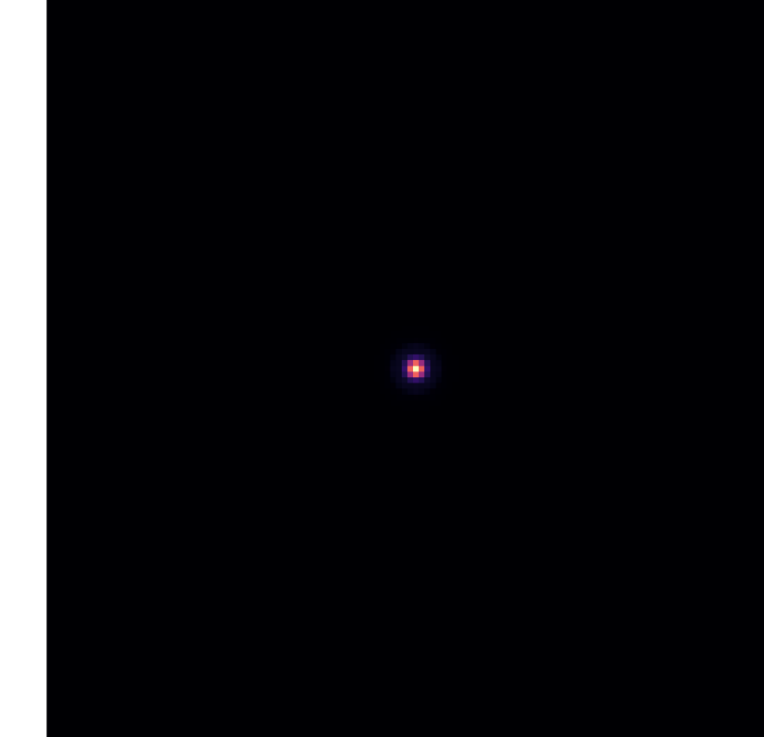
$$P(r) = \exp \left[\underbrace{j2C_d r^2 / R^2}_{\text{Defocus}} + \underbrace{jC_s r^4 / R^4}_{\text{Spherical Aberration}} \right] \approx \begin{cases} \exp[j\alpha_1 r^2] & 0 \leq r \leq r_1 \\ \vdots & \vdots \\ \exp[j\alpha_M r^2] & r_{M-1} \leq r \leq r_M \end{cases}$$

PSF Accuracy

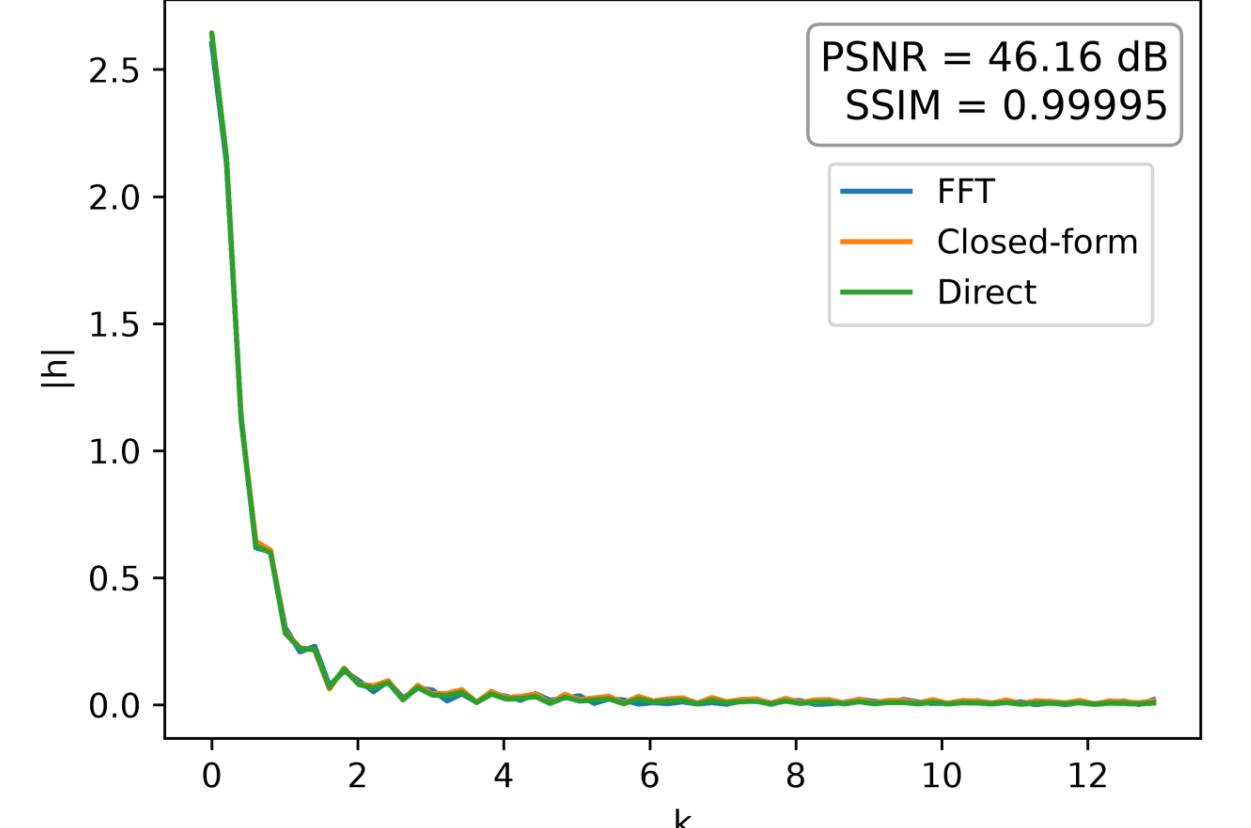
Numerical Reference (FFT)
 $C_d = 1, C_s = 0$



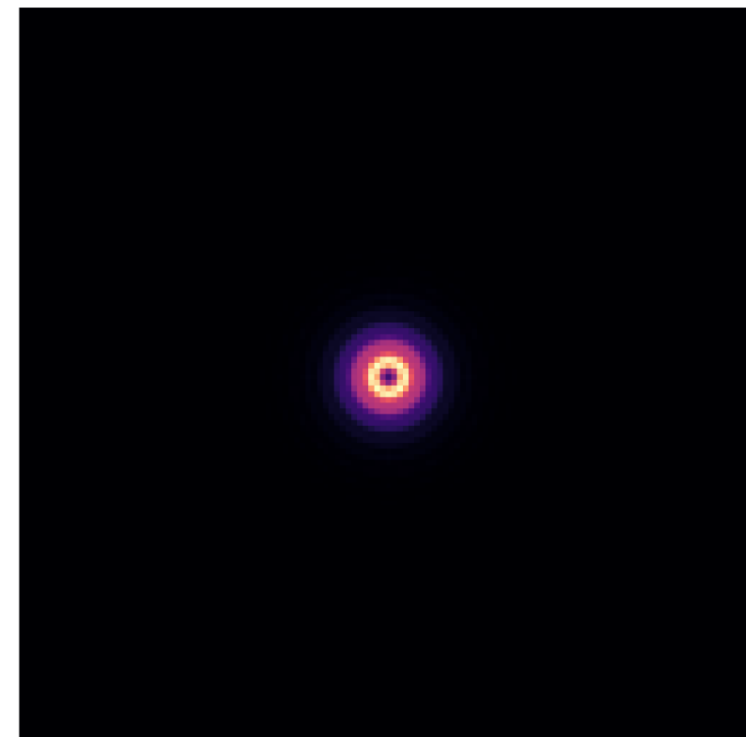
Closed-Form Approximation
 $C_d = 1, C_s = 0$



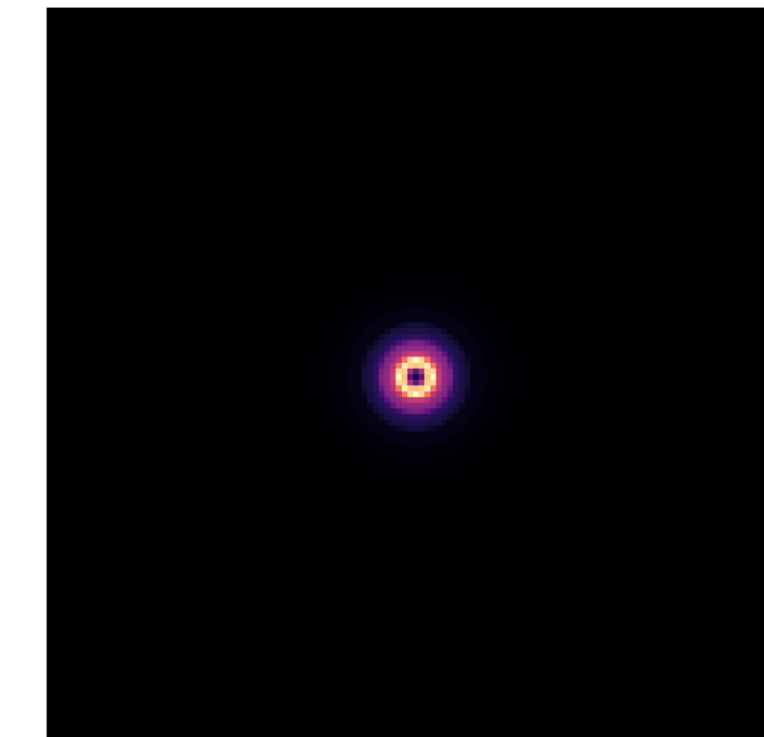
Radial Slice Comparison $C_s = 0, C_d = 1$



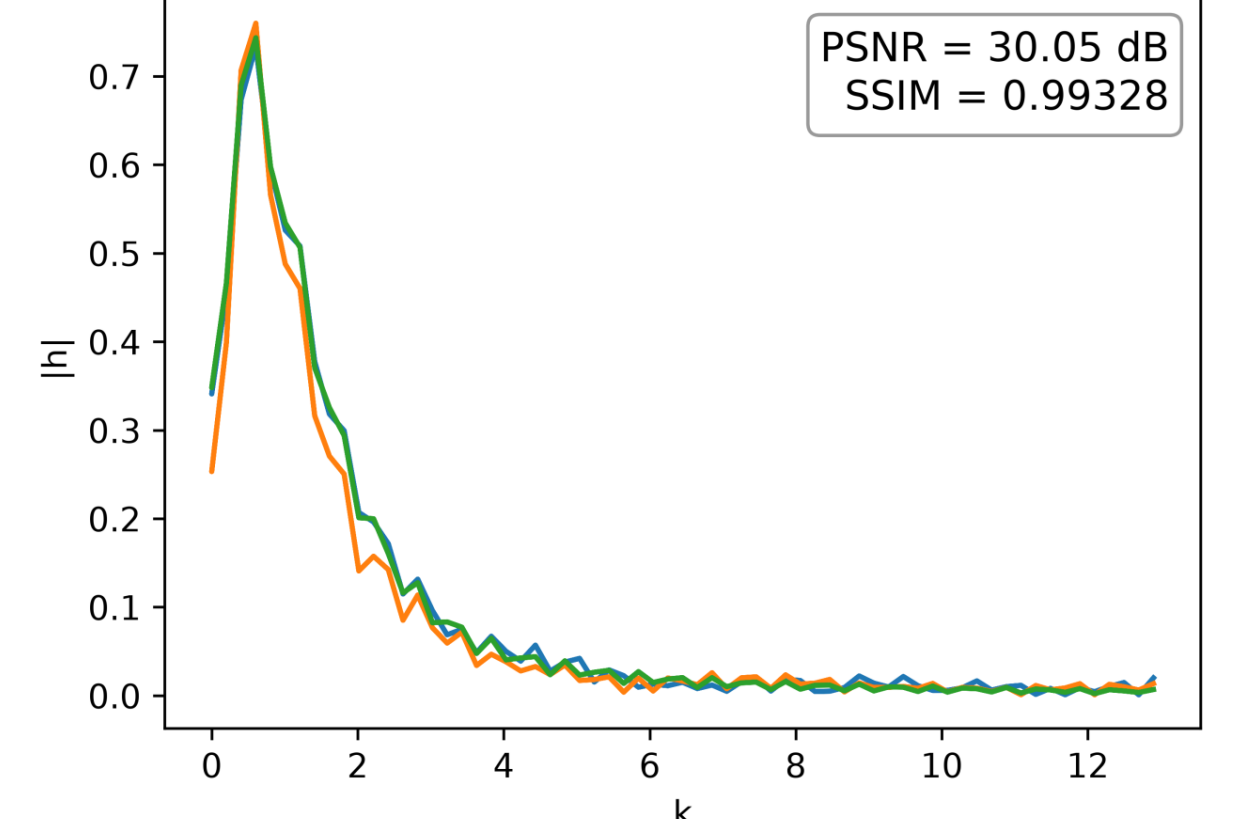
Numerical Reference (FFT)
 $C_d = 2.0, C_s = 2.0$



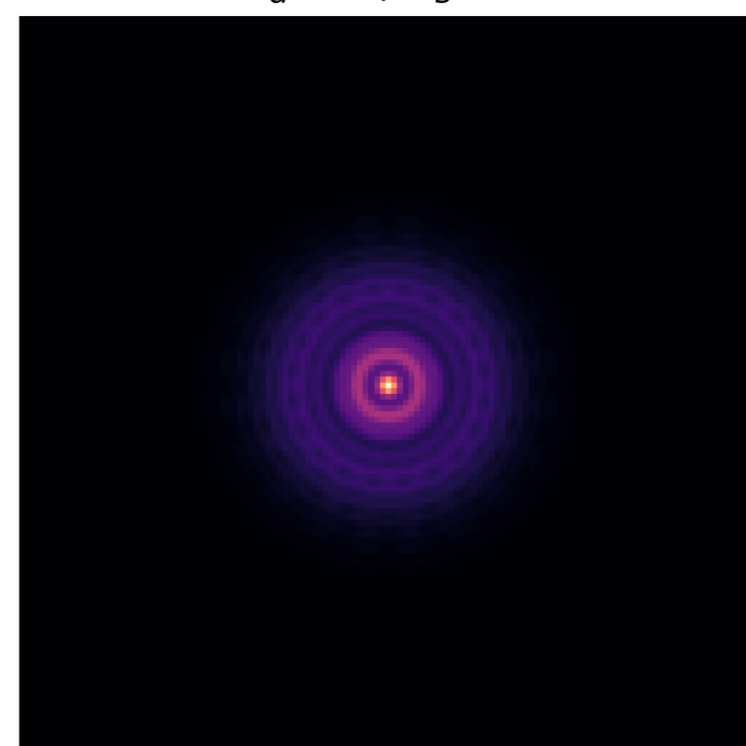
Closed-Form Approximation
 $C_d = 2.0, C_s = 2.0$



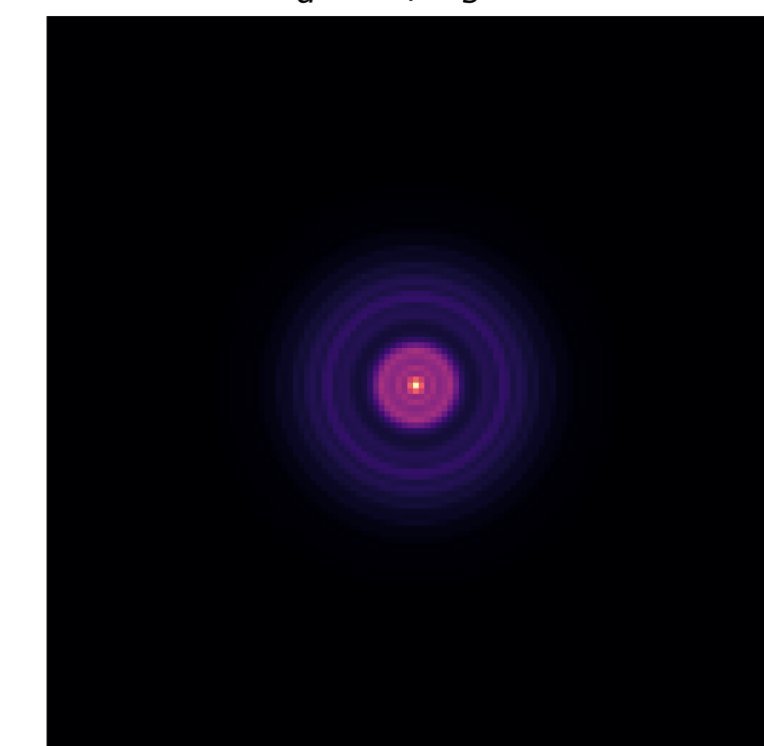
Radial Slice Comparison $C_s = 2.0, C_d = 2.0$



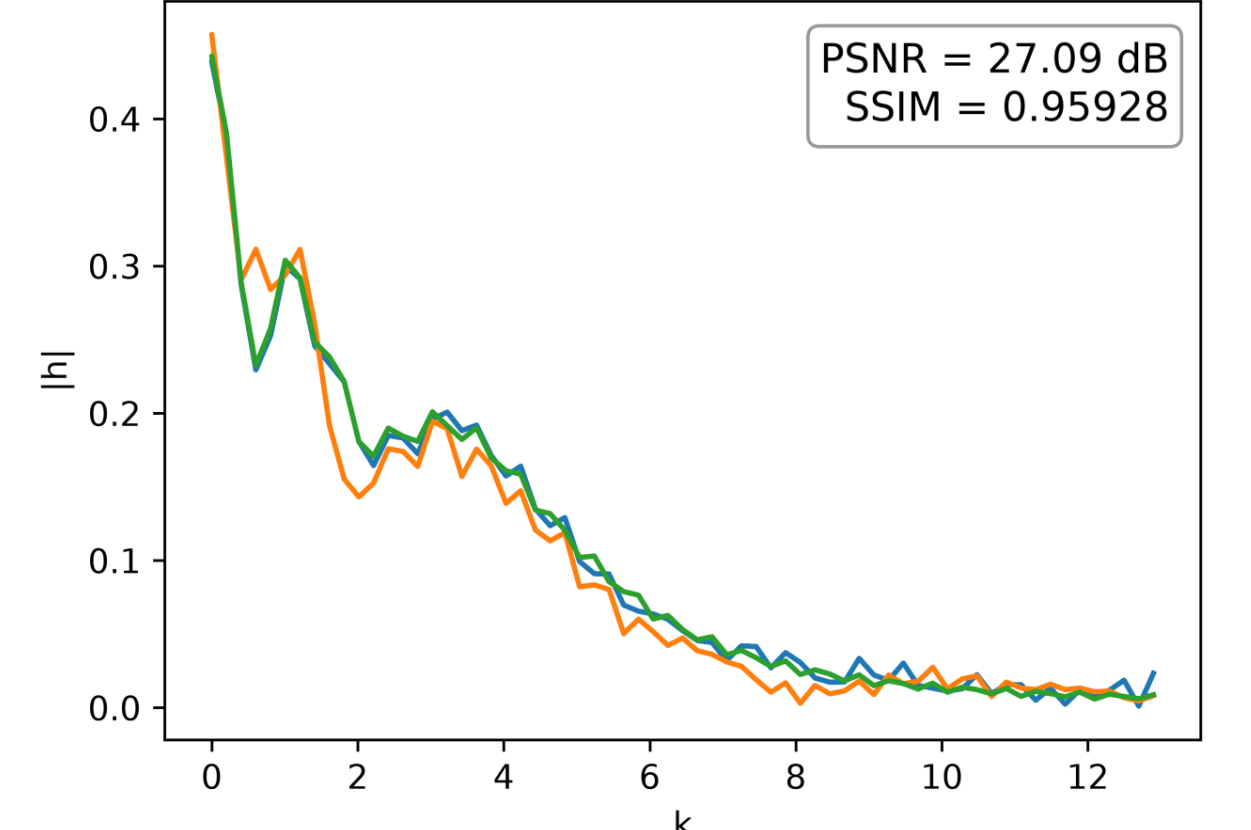
Numerical Reference (FFT)
 $C_d = 5, C_s = 5$



Closed-Form Approximation
 $C_d = 5, C_s = 5$



Radial Slice Comparison $C_s = 5, C_d = 5$



Depth-of-field Synthesis

